

EE/CE 354/361: Introduction to Probability and Statistics

Homework Assignment 2: Discrete Random Variables

Instructions

- **Attempt all questions.** Only **7 randomly chosen** questions will be graded.
- Show all working clearly.
- State assumptions where necessary.
- Total Points: **70**

Q1 (10 points)

A fair coin is tossed three times. Let X be the number of heads obtained.

- Find the PMF of X .
- Plot the PMF.
- Compute $P(X = 2)$ and $P(X \geq 1)$.

Q2 (10 points)

A discrete random variable X takes values 0, 1, 2, 3 with

$$p_X(k) = P(X = k) = c(2k + 1), \quad k = 0, 1, 2, 3.$$

- Determine c .
- Compute $P(X > 1)$.
- Find the Mean of X .
- Find the Variance of X .

Q3 (10 points)

A coin is tossed repeatedly until a Head occurs. The probability of obtaining a Head on a single toss is p . Let the discrete random variable X denote the number of tosses required for a Head to appear for the first time.

- What is the PMF of the random variable X .
- Show that $E[X] = \frac{1}{p}$.
- Find $\text{Var}(X)$.

Q4 (10 points)

From the Bernoulli, Binomial, Geometric, Uniform, and Poisson random variables, which one will be the most appropriate choice to model the following:

- a) Number of games Karachi Kings will win in the 2026 edition of Pakistan Super League
- b) Outcome of rolling a fair 6-sided die
- c) Number of balls that will be bowled by a new player in international cricket before getting his/her first wicket.

Q5 (10 points)

- a) One out of every eight phone calls to your house is a telemarketer. Assume that the telemarketers' calls are independent. Let X denote the number of telemarketers during the next three calls. What is the PMF of X ?
- b) One out of every eight phone calls to your house is a telemarketer. Assume that the telemarketers' calls are independent. Let X denote the number of phone calls until the telemarketer calls (If the second call is a telemarketer, $X = 2$ and so on). What is the probability that $X \geq 3$?

Q6 (10 points)

Let the random variable X be the outcome of rolling a fair 6-sided die.

- a) What is $E[X^2 + 1]$?
- b) What is $E[2X + 1]$?

Q7 (10 points)

Consider the following random variables X & Y :

X = Magnitude of the difference between the outcomes of two rolls of a fair 3-sided die.

- (If the first roll results in 1 and second roll results in 3, the value taken by X is 2)
- (If the first roll results in 3 and second roll results in 1, the value taken by X is 2)

$Y = 2X$

- a) Calculate and plot the PMF of X .
- b) Calculate and plot the PMF of Y .
- c) Calculate Mean of Y .
- d) Calculate Variance of Y .

Q8 (10 points)

Suppose you roll a fair six-sided Red die and a fair tetrahedral Green die. Consider the following two random variables X & Y :

X = number shown on the fair six-sided Red die

$$Y = \begin{cases} 0 & \text{if the two dice show the same number,} \\ 1 & \text{if the number on the green tetrahedral die is bigger than the number on the red die,} \\ 2 & \text{if the number on the red six-sided die is bigger than the number on the green die.} \end{cases}$$

- a) What is $p_{X,Y}(x,y)$? (You can write it in table form).
- b) What is $p_Y(y)$?
- c) What is $E[X]$?
- d) What is $p_{X|Y}(x|2)$?

Q9 (10 points)

A Random Variable Z_n is defined as $Z_n = \min(X_1, X_2, \dots, X_n)$, where X_i 's are outcomes of independent rolls of fair 3-sided die. As an example, $Z_2 = \min(X_1, X_2)$ - meaning Z_2 is the minimum of the outcome of two independent rolls of fair 3-sided die X_1 and X_2 .

- a) What is the mean of Z_2 ?
- b) Let's define a random variable Y as $Y = 2Z_1 + Z_2$. What is the mean of Y ?

Q10 (10 points)

Consider four independent rolls of a fair 6-sided die. Let X be the number of 1's and let Y be the number of 2's obtained. What is the joint PMF of X and Y ? (You can give your answer in tabular form).

Q11 (10 points)

Let X be the number of heads obtained when three fair coins are tossed. Given that at least one head occurs, find:

- a) The conditional PMF of X given that at least one head occurs, i.e. $p_{X|A}(x)$, where $A = \{X \geq 1\}$.
- b) The conditional expectation $E[X | A]$.

Q12 (10 points)

Let's assume that 40% of current HU students belong to AHSS and 60% belong to DSSE. Let X be the random variable that a randomly selected HU student belongs to DSSE. (X takes on value 1 if the student belongs to DSSE and 0 otherwise.) Let Y be the random variable that captures the CGPA of a randomly selected HU student. You have been provided the following information:

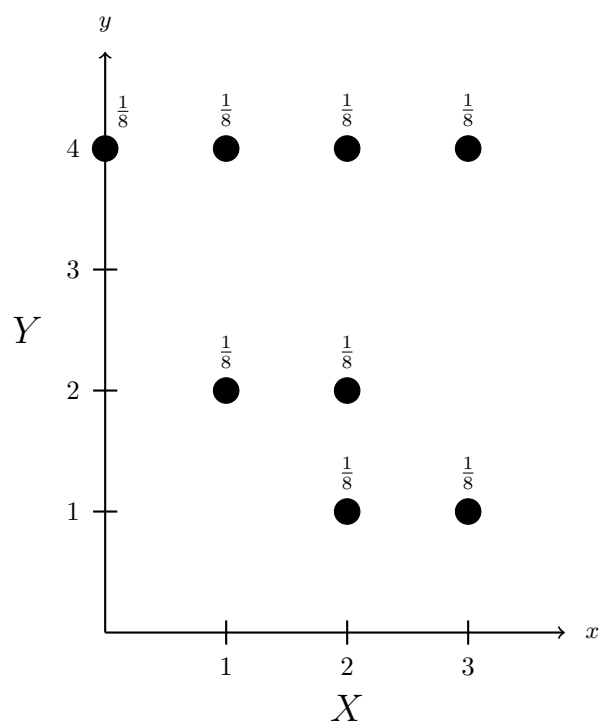
$$E[Y|X = 1] = 3.0 \quad E[Y|X = 0] = 3.2$$

Consider the following exercise: Stand at the main entrance gate of HU, ask the CGPA of first 100 students, and calculate their average CGPA. Before conducting the above-mentioned exercise, what will be your best guess for the average CGPA obtained through the exercise? Justify your answer by pointing out the relevant laws/theorems discussed in this course?

(Note: No points will be awarded for an answer without a valid justification and identification of the appropriate theorem/laws discussed in the course).

Q13 (10 points)

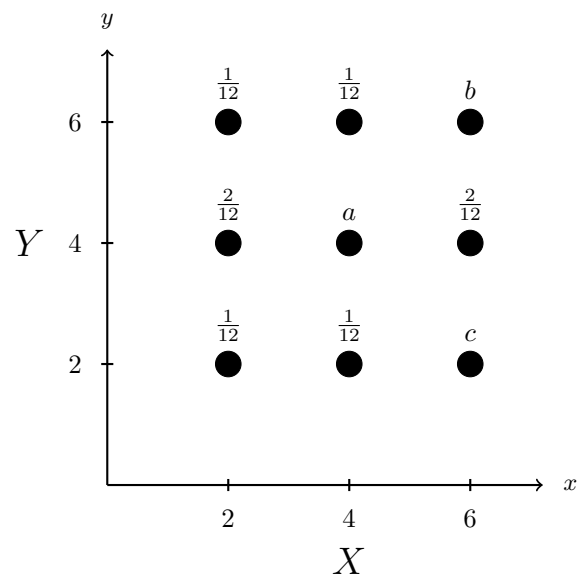
Consider the following joint PMF of two random variables X and Y :



- Are X and Y independent?
- Draw the following: $p_{X|Y}(x|1)$, $p_{Y|X}(y|1)$.
- What is $E[X|Y = 4]$?
- Let $R = \max(X, Y)$, what is the mean of R ?

Q14 (10 points)

Let X and Y be two discrete random variables with their joint PMF shown below:



- Calculate $p_X(2)$.
- Calculate $E[Y|X=2]$
- Determine the values of a , b , and c , under which X and Y are independent random variables. Justify your answer.